Chapter 1 Review Name_____

Determine whether the equation defines y as a function of x.

1. $y = -9x^2 + 5x - 2$
2. $x = y^2 + 6$

Find the domain of the given function.

3. $\sqrt{7 - x}$
4. $\dfrac{x + 9}{(x + 4)(x - 8)}$

Find the range of the function.

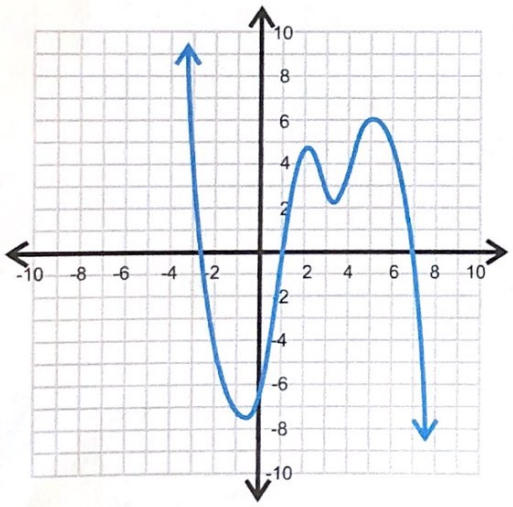
5. $f(x) = (x - 1)^2 + 4$

Determine where each function is continuous or discontinuous. If there is a discontinuity, tell whether it is removable or non-removable.

6. $f(x) = \dfrac{(x - 2)(x + 5)}{x^2 - 4}$
7. $f(x) = \dfrac{x^2 - 3x}{x}$

8. Use the graph below to estimate the local maximum and local minimum.

9. Determine the intervals on which the function is increasing, decreasing, and constant.



Determine if the function is bounded above, bounded below, bounded or unbounded.

10. $y = x^2 + 5$

11. $\sqrt{9 - x^2}$

12. $y = x^3$

Determine whether the function is even, odd, or neither.

13. $y = 3x^2 - 3$

14. $f(x) = 8 + 5x + x^2$

Perform the requested operation or operations given *f(x) = 7x + 14; g(x) = 4x - 1*

15. $(f + g)(x)$

16. $(f - g)(x)$

17. $(f \cdot g)(x)$

18. $\left(\frac{f}{g}\right)(x)$

19. $(f \circ g)(x)$

20. $(g \circ f)(x)$

Find the inverse of each function.

21. $f(x) = x^3 + 7$

22. $f(x) = \frac{-2x - 7}{6x + 8}$

Describe how the parent graph $y = x^2$ can be transformed to the graph of the given equation.

23. $y = -2x^2 + 9$

24. $y = (x - 4)^2 + 6$